

Midterm 1 Practice

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Short Answer Questions

1 Define the Impulse Response

The impulse response, $h(n)$, is defined as the output of a system when its input is $\delta(n)$.

2 Consider the signal $x(n) = \cos(\frac{2\pi}{3}n)$

2.a What is the Period of $x(n)$?

The for a discrete periodic function, its period is defined as $P \in \mathbb{N}$ such that $x(n) = x(n+P) \forall n \in \mathbb{N}$. Using this fact, we find:

$$\begin{aligned} x(n) = x(n+P) &\implies \\ \cos(\frac{2\pi}{3}n) &= \cos(\frac{2\pi}{3}(n+P)) \\ &= \cos(\frac{2\pi}{3}n + \frac{2\pi}{3}P) \end{aligned}$$

Therefore, since cosine is 2π periodic, the smallest value of P which makes $\frac{2\pi}{3}P$ a multiple of 2π is 3. Therefore, the period of $x(n)$ is 3.

2.b What is the Discrete Frequency of $x(n)$?

The discrete frequency f of a P -periodic discrete system is given by $f = \frac{1}{P}$. Therefore, the discrete frequency of $x(n)$ is $\frac{1}{3}$.

3 What is the Frequency Response ($H(\omega)$) of the system $y'' + y = x$?

We know that the frequency response $H(\omega)$ of a continuous system is given by $H(\omega)e^{i\omega t} = y(t)$ when $x(t) = e^{i\omega t}$. Using this, we can find that:

$$\begin{aligned}y'' + y &= x \implies \\(H(\omega)e^{i\omega t})'' + H(\omega)e^{i\omega t} &= e^{i\omega t} \\-\omega^2 H(\omega) + H(\omega) &= 1 \\H(\omega)(1 - \omega^2) &= 1 \\H(\omega) &= \frac{1}{1 - \omega^2}\end{aligned}$$

Therefore, the frequency response of the system is $H(\omega) = \frac{1}{1 - \omega^2}$.

4 What are the units for:

4.a The period of a signal?

The units for period are $\frac{\text{samples}}{\text{cycle}}$.

4.b The discrete frequency of a signal?

The units for discrete frequency are $\frac{\text{periods}}{\text{sample}}$.

State Space

5 Given the difference equation $y(n) = y(n-2) - x(n-2)$, determine the (A, B, C, D) representation of the system.

5.I Find $\overrightarrow{S(n)}$

Let:

$$s_1(n) = x(n-2)$$

$$s_2(n) = x(n-1)$$

$$s_3(n) = y(n-2)$$

$$s_4(n) = y(n-1)$$

Then:

$$s_1(n+1) = x(n-1) = s_2(n)$$

$$s_2(n+1) = x(n)$$

$$s_3(n+1) = y(n-1) = s_4(n)$$

$$s_4(n+1) = y(n) = -x(n-2) + y(n-2) = -s_1(n) + s_3(n)$$

Thus, let $\overrightarrow{S(n)} = \begin{pmatrix} s_1(n) \\ s_2(n) \\ s_3(n) \\ s_4(n) \end{pmatrix}$.

5.II Find $\overrightarrow{S(n+1)}$

$$\overrightarrow{S(n+1)} = \begin{pmatrix} s_1(n+1) \\ s_2(n+1) \\ s_3(n+1) \\ s_4(n+1) \end{pmatrix}$$

$$= \begin{pmatrix} s_2(n) \\ x(n) \\ s_4(n) \\ -s_1(n) + s_3(n) \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & 1 & 0 \end{pmatrix} \overrightarrow{S(n)} + \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} x(n)$$

Therefore, our $\mathbf{A}$ and $\mathbf{B}$ matrices are $\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & 1 & 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$ respectively.

5.III Find $y(n)$

$$\begin{aligned} y(n) &= -x(n-2) + y(n-2) \\ &= -s_1(n) + s_3(n) \\ &= \begin{pmatrix} -1 & 0 & 1 & 0 \end{pmatrix} \overrightarrow{S(n)} + \begin{pmatrix} 0 \end{pmatrix} x(n) \end{aligned}$$

Therefore, our $\mathbf{C}$ and $\mathbf{D}$ matrices are $\begin{pmatrix} -1 & 0 & 1 & 0 \end{pmatrix}$ and $\begin{pmatrix} 1 \end{pmatrix}$ respectively.

Putting it all together, our system can be represented in state space form as:

$$\overrightarrow{S(n+1)} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & 1 & 0 \end{pmatrix} \overrightarrow{S(n)} + \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} x(n)$$

$$y(n) = \begin{pmatrix} -1 & 0 & 1 & 0 \end{pmatrix} \overrightarrow{S(n)} + \begin{pmatrix} 0 \end{pmatrix} x(n)$$

6 Given the matrices:

$$A = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, B = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, C = \begin{pmatrix} 1 & 1 \end{pmatrix}, D = \begin{pmatrix} 1 \end{pmatrix}$$

Compute the zero state output to the input $x(n) = \delta(n) + \delta(n-1)$ (Find $y(n)$ for $n = 0, 1, 2, 3$)

Current State ($\overrightarrow{S(n)}$)

Current Output ($y(n)$)

$$\overrightarrow{S(0)} = \vec{0}$$

$$y(0) = \begin{pmatrix} 1 & 1 \end{pmatrix} \overrightarrow{S(0)} + \begin{pmatrix} 1 \end{pmatrix} x(0) = 1$$

$$\overrightarrow{S(1)} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \overrightarrow{S(0)} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} x(0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$y(1) = \begin{pmatrix} 1 & 1 \end{pmatrix} \overrightarrow{S(1)} + \begin{pmatrix} 1 \end{pmatrix} x(1) = 2$$

$$\overrightarrow{S(2)} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \overrightarrow{S(1)} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} x(1) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$y(2) = \begin{pmatrix} 1 & 1 \end{pmatrix} \overrightarrow{S(2)} + \begin{pmatrix} 1 \end{pmatrix} x(2) = 2$$

$$\overrightarrow{S(3)} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \overrightarrow{S(2)} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} x(2) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$y(3) = \begin{pmatrix} 1 & 1 \end{pmatrix} \overrightarrow{S(3)} + \begin{pmatrix} 1 \end{pmatrix} x(3) = 1$$

7 Given the state-space equations:

$$\begin{aligned}\overrightarrow{S(n+1)} &= \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \overrightarrow{S(n)} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} x(n) \\ y(n) &= \begin{pmatrix} 1 & 1 \end{pmatrix} \overrightarrow{S(n)} + \begin{pmatrix} 1 \end{pmatrix} x(n)\end{aligned}$$

Compute the zero-state output for the input $x(n) = \delta(n)$. (Find $y(n)$ for $n = 0, 1, 2, 3, 4$)

Current State ($\overrightarrow{S(n)}$)

Current Output ($y(n)$)

$$\overrightarrow{S(0)} = \vec{0}$$

$$y(0) = \begin{pmatrix} 1 & 1 \end{pmatrix} \overrightarrow{S(0)} + \begin{pmatrix} 1 \end{pmatrix} x(0) = 1$$

$$\overrightarrow{S(1)} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \overrightarrow{S(0)} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} x(0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$y(1) = \begin{pmatrix} 1 & 1 \end{pmatrix} \overrightarrow{S(1)} + \begin{pmatrix} 1 \end{pmatrix} x(1) = 1$$

$$\overrightarrow{S(2)} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \overrightarrow{S(1)} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} x(1) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$y(2) = \begin{pmatrix} 1 & 1 \end{pmatrix} \overrightarrow{S(2)} + \begin{pmatrix} 1 \end{pmatrix} x(2) = 2$$

$$\overrightarrow{S(3)} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \overrightarrow{S(2)} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} x(2) = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$y(3) = \begin{pmatrix} 1 & 1 \end{pmatrix} \overrightarrow{S(3)} + \begin{pmatrix} 1 \end{pmatrix} x(3) = 3$$

$$\overrightarrow{S(4)} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \overrightarrow{S(3)} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} x(3) = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

$$y(4) = \begin{pmatrix} 1 & 1 \end{pmatrix} \overrightarrow{S(4)} + \begin{pmatrix} 1 \end{pmatrix} x(4) = 5$$

Convolution

- 8 An LTI system is given by its frequency response $h(n) = \delta(n) + \delta(n - 2)$. Determine the output of the system to the input $x(n) = \delta(n) + \delta(n - 2) + \delta(n - 4)$**

Although we could use the convolutional equation here, since the system is given to be LTI, it is much easier to just take the linear combination of $h(n)$ and $x(n)$ directly. Thus:

$$\begin{aligned} y(n) &= h(n) + h(n - 2) + h(n - 4) \\ &= \delta(n) + \delta(n - 2) + \delta(n - 2) + \delta(n - 4) + \delta(n - 4) + \delta(n - 6) \\ &= \delta(n) + 2\delta(n - 2) + 2\delta(n - 4) + \delta(n - 6) \end{aligned}$$

Therefore, the output of the system for the input $x(n) = \delta(n) + \delta(n - 2) + \delta(n - 4)$ is given by:

$$y(n) = \delta(n) + 2\delta(n - 2) + 2\delta(n - 4) + \delta(n - 6)$$

OR

$$y(n) = \begin{cases} 1, & \text{if } n = 0, 6 \\ 2, & \text{if } n = 2, 4 \\ 0, & \text{otherwise} \end{cases}$$

The Frequency Response

9 Compute the frequency response ($H(\omega)$) of the system $y(n) + y(n-1) = 2x(n) - 5x(n-2)$.

Using the familiar fact that $y(n) = H(\omega)e^{i\omega n}$ when $x(n) = e^{i\omega n}$, we can find that:

$$\begin{aligned}
 y(n) + y(n-1) &= 2x(n) - 5x(n-2) \implies \\
 H(\omega)e^{i\omega n} + H(\omega)e^{i\omega(n-1)} &= 2e^{i\omega n} - 5e^{i\omega(n-2)} \\
 H(\omega)e^{i\omega n} + H(\omega)e^{i\omega n-i\omega} &= 2e^{i\omega n} - 5e^{i\omega n-2i\omega} \\
 H(\omega)e^{i\omega n} + H(\omega)e^{i\omega n}e^{-i\omega} &= 2e^{i\omega n} - 5e^{i\omega n}e^{-2i\omega} \\
 H(\omega) + H(\omega)e^{-i\omega} &= 2 - 5e^{-2i\omega} \\
 H(\omega)(1 + e^{-i\omega}) &= 2 - 5e^{-2i\omega} \\
 H(\omega) &= \frac{2 - 5e^{-2i\omega}}{1 + e^{-i\omega}}
 \end{aligned}$$

Therefore, the frequency response of the system is given by $H(\omega) = \frac{2-5e^{-2i\omega}}{1+e^{-i\omega}}$.

10 The frequency response of a system is given by $H(\omega) = \cos(\omega)$. Compute the output of the system to the signal $x(n) = 2 + \cos(\frac{\pi}{2}n + \frac{\pi}{4}) + \sin(\pi n + \frac{3\pi}{2})$.

10.I Find the frequency of each term of $x(n)$

Terms of $x(n)$	Angular Frequency (ω)
2	0
$\cos(\frac{\pi}{2}n + \frac{\pi}{4})$	$\frac{\pi}{2}$
$\sin(\pi n + \frac{3\pi}{2})$	π

10.II Find $|H(\omega)|$ and $\arg(H(\omega))$ for each term of $x(n)$

Terms of $x(n)$	Angular Frequency (ω)	Frequency Response ($H(\omega)$)	Gain ($ H(\omega) $)	Phase Shift ($\arg(H(\omega))$)
2	0	1	1	0
$\cos(\frac{\pi}{2}n + \frac{\pi}{4})$	$\frac{\pi}{2}$	0	0	—
$\sin(\pi n + \frac{3\pi}{2})$	π	-1	1	0

10.III Compute $y(n)$ using the cosine formula

THIS ANSWER IS WRONG, I NEED TO REACH OUT TO A TA TO DETERMINE WHY

Note here that since $\arg(H(\pi))$ does not exist, we do not include it in our computation.

$$\begin{aligned}y(n) &= |H(\omega_1)| \cos(\omega_1 n + \arg(H(\omega_1))) + |H(\omega_2)| \cos(\omega_2 n + \arg(H(\omega_2))) \\&= \cos((0)n + 0) + \cos((\pi)n + 0) \\&= 1 + \cos((\pi)n)\end{aligned}$$

Therefore, the output of a system with frequency response $H(\omega)$ for the input $x(n) = 2 + \cos(\frac{\pi}{2}n + \frac{\pi}{4}) + \sin(\pi n + \frac{3\pi}{2})$ is given by $y(n) = 1 + \cos((\pi)n)$.

11 A system's frequency response is given by $H(\omega) = \frac{1+e^{-i\omega}}{e^{-i\omega}}$. Compute the output of the system for the input signal $x(n) = \cos(\frac{\pi}{2}n + \frac{\pi}{4}) + \sin(\pi n + \frac{3\pi}{2})$

FIX BASED ON FEEDBACK!!!!

Impulse Response

12 Find a closed-form expression for the impulse response of the system $y(n) = x(n) - 2x(n-1) + 3x(n-3)$

Since the system is FIR, we can simply substitute $x(n)$ for $\delta(n)$ to find that:

$$\begin{aligned}y(n) &= x(n) - 2x(n-1) + 3x(n-3) \\ &= \delta(n) - 2\delta(n-1) + 3\delta(n-3)\end{aligned}$$

Therefore, the impulse response of the system is given by $h(n) = \delta(n) - 2\delta(n-1) + 3\delta(n-3)$

Complex Numbers

13 Express the following numbers as complex exponential

13.a $z = 1 + i$

$$|z| = \sqrt{(1)^2 + (1)^2} = \sqrt{2}$$
$$\arg(z) = \frac{\pi}{4} = \frac{-7\pi}{4}$$

Therefore, $z = 1 + i = \sqrt{2}e^{i\frac{-7\pi}{4}}$.

13.b $z = -1$

$$|z| = \sqrt{(-1)^2 + (0)^2} = 1$$
$$\arg(z) = \pi = -\pi$$

Therefore, $z = -1 = e^{i-\pi}$.

13.c $z = -i$

$$|z| = \sqrt{(0)^2 + (-1)^2} = 1$$
$$\arg(z) = \frac{3\pi}{2} = \frac{-\pi}{2}$$

Therefore, $z = -i = e^{i\frac{-\pi}{2}}$.

13.d $z = -1 - i$

$$|z| = \sqrt{(-1)^2 + (1)^2} = \sqrt{2}$$
$$\arg(z) = \frac{5\pi}{4} = \frac{-3\pi}{4}$$

Therefore, $z = -1 - i = e^{i\frac{-3\pi}{4}}$.

14 Find the argument of the following complex numbers

14.a $z = \frac{1}{1+i}$

$$\begin{aligned} z &= \frac{1}{1+i} \\ &= \frac{1}{1+i} \cdot \frac{1-i}{1-i} \\ &= \frac{1-i}{(1)^2 - (i)^2} \\ &= \frac{1-i}{2} \end{aligned}$$

Therefore, $\arg(z) = \arg(\frac{1-i}{2}) = \arg(\frac{1-i}{2}) = \frac{-\pi}{4}$.

14.b $z = -2i$

Therefore, $\arg(z) = \arg(-2i) = \frac{-\pi}{2}$.

14.c 2

Therefore, $\arg(z) = \arg(2) = 0$.

Discrete Fourier Series

15 Find the discrete Fourier Series of the signal $x(n) = 1 + \cos(\frac{1}{2}\pi n)$ where $N = 4$

To begin, we first compute $x(0), x(1), x(2)$, and $x(3)$.

$$x(0) = 1 + \cos\left(\frac{1}{2}\pi(0)\right) = 1 + 1 = 2$$

$$x(1) = 1 + \cos\left(\frac{1}{2}\pi(1)\right) = 1 + 0 = 1$$

$$x(2) = 1 + \cos\left(\frac{1}{2}\pi(2)\right) = 1 + -1 = 0$$

$$x(3) = 1 + \cos\left(\frac{1}{2}\pi(3)\right) = 1 + 0 = 1$$

Since the fundamental frequency of 1 and $\cos(\frac{1}{2}\pi n)$ is 1 and $\frac{\pi}{2}$ respectively, the fundamental frequency (ω_0) of the signal $x(n)$ is $\frac{\pi}{n}$.

Using this, we can compute $X(0), X(1), X(2)$, and $X(3)$.

$$X(0) = \frac{1}{N} \sum_{n=0}^{N-1} (x(n)e^{-i\omega_0(0)n}) = \frac{1}{4} \sum_{n=0}^3 (x(n)) = \frac{1}{4}(2 + 0 + 1 + 1) = 1$$

$$X(1) = \frac{1}{N} \sum_{n=0}^{N-1} (x(n)e^{-i\omega_0(1)n}) = \frac{1}{4} \sum_{n=0}^3 (x(n)e^{-i\frac{\pi}{2}n}) = \frac{1}{4}(2 + e^{-i\frac{\pi}{2}} + e^{-3i\frac{\pi}{2}}) = \frac{1}{4}(2 - i + i) = \frac{1}{2}$$

$$X(2) = \frac{1}{N} \sum_{n=0}^{N-1} (x(n)e^{-i\omega_0(2)n}) = \frac{1}{4} \sum_{n=0}^3 (x(n)e^{-i\pi n}) = \frac{1}{4}(2 + e^{-i\pi} + e^{-3i\pi}) = \frac{1}{4}(2 - 1 - 1) = 0$$

$$X(3) = \frac{1}{N} \sum_{n=0}^{N-1} (x(n)e^{-i\omega_0(3)n}) = \frac{1}{4} \sum_{n=0}^3 (x(n)e^{-i\frac{3\pi}{2}n}) = \frac{1}{4}(2 + e^{-i\frac{3\pi}{2}} + e^{-i\frac{9\pi}{2}}) = \frac{1}{4}(2 + i - i) = \frac{1}{2}$$

Filter Design

16 Give the difference equation of a causal, minimum order, discrete and real valued system that removed any frequency of 0 and $\frac{\pi}{2}$ and passes signals with a normalized frequency of π with a gain of 1

To create such a system, we must first find an appropriate frequency response for the system.

16.I Find an appropriate $H(\omega)$ for the system

We know that:

$$H(0) = 0, H\left(\frac{\pi}{2}\right) = 0, |H(\pi)| = 1$$

Therefore, by factoring to ensure our filter has the correct zeros (and ensuring complex conjugate symmetry), we can arrive at a prototype $\tilde{H}(\omega)$ of the system's frequency response:

$$\begin{aligned}\tilde{H}(\omega) &= (e^{-i\omega} - e^{-i\omega(0)})(e^{-i\omega} - e^{-i\frac{\pi}{2}})(e^{-i\omega} - e^{i\frac{\pi}{2}}) \\ &= (e^{-i\omega} - 1)(e^{-i\omega} - i)(e^{-i\omega} + i) \\ &= e^{-3i\omega} - e^{-2i\omega} + e^{-i\omega} - 1\end{aligned}$$

To arrive at $H(\omega)$, we want to normalize our prototype filter by ensuring it has a gain of 1 when $\omega = \pi$. Plugging $\omega = \pi$ into our prototype filter gives us:

$$\begin{aligned}\tilde{H}(\pi) &= e^{-3i\pi} - e^{-2i\pi} + e^{-i\pi} - 1 \\ \tilde{H}(\pi) &= 4 \implies \\ |\tilde{H}(\pi)| &= 4\end{aligned}$$

Therefore, $H(\omega) = \frac{-1}{4}\tilde{H}(\omega) = \frac{1}{4} - \frac{1}{4}e^{-i\omega} + \frac{1}{4}e^{-2i\omega} - \frac{1}{4}e^{-3i\omega}$ gives us our system's frequency response with the correct gain at π . Note here that we could choose to multiply by either $\frac{1}{4}$ or $\frac{-1}{4}$, which has no effect on our gain, I simply chose $\frac{-1}{4}$ because the final solution provided does this.

16.II Find the difference equation

Using the identity $y(n) = H(\omega)e^{i\omega n}$ when $x(n) = e^{i\omega n}$, we can multiply both sides of our original equation to arrive at a difference equation for the system.

$$\begin{aligned}H(\omega) &= \frac{1}{4} - \frac{1}{4}e^{-i\omega} + \frac{1}{4}e^{-2i\omega} - \frac{1}{4}e^{-3i\omega} \implies \\ H(\omega)e^{i\omega n} &= \frac{1}{4}e^{i\omega n} - \frac{1}{4}e^{-i\omega}e^{i\omega n} + \frac{1}{4}e^{-2i\omega}e^{i\omega n} - \frac{1}{4}e^{-3i\omega}e^{i\omega n} \implies \\ y(n) &= \frac{1}{4}e^{i\omega n} - \frac{1}{4}e^{-i\omega+i\omega n} + \frac{1}{4}e^{-2i\omega+i\omega n} - \frac{1}{4}e^{-3i\omega+i\omega n} \\ &= \frac{1}{4}e^{i\omega n} - \frac{1}{4}e^{i\omega(n-1)} + \frac{1}{4}e^{i\omega(n-2)} - \frac{1}{4}e^{i\omega(n-3)} \\ &= \frac{1}{4}x(n) - \frac{1}{4}x(n-1) + \frac{1}{4}x(n-2) - \frac{1}{4}x(n-3)\end{aligned}$$

Therefore, the difference equation of our system is given by $y(n) = \frac{1}{4}x(n) - \frac{1}{4}x(n-1) + \frac{1}{4}x(n-2) - \frac{1}{4}x(n-3)$.